Target Driven Guidance for Extreme Weather Storylines

Master's Thesis

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Abstract

An increasingly important class of machine-learning weather models is formed by generative probabilistic models. By learning the conditional distribution of future atmospheric states, they naturally enable the generation of ensemble forecasts through repeated sampling. Their generative formulation can also be exploited to steer the sampling process toward desired outcomes. While *guidance* is widely used in image generation, it remains comparatively unexplored in the weather domain. This thesis investigates whether generative weather models can be deliberately steered to produce plausible extreme-weather trajectories.

Existing guidance approaches commonly prescribe the strength of the intervention and subsequently observe how the generated sample changes. We invert this procedure by specifying the desired outcome first and determining the guidance strength required to realize it separately. We formulate the resulting approach in two stages, *Specify* and *Guide*. In the first stage, we specify the scenario through a target variable, a spatial region, and a trajectory describing how the corresponding regional quantity should evolve over the forecast horizon. In the second stage, we steer the generative dynamics toward these targets using gradient-based guidance.

To realize these targets, we distribute guidance over the generative process according to a prescribed relative strength profile and determine the required scalar gain independently at each weather step through a root-finding procedure.

We investigate this approach using ArchesWeather, a medium-range probabilistic weather model based on flow matching. As a case study, we generate regional high-temperature scenarios by prescribing trajectories for the regional mean surface temperature. We systematically study how different distributions of guidance across the generative process affect target realization and the properties of the resulting trajectories.

We find that concentrating guidance at the second flow step consistently yields the most favorable results across different evaluation perspectives.

Our results show how guidance can extend generative probabilistic weather models beyond sampling likely forecast trajectories toward the controlled generation of prescribed weather scenarios, providing a means to explore low-probability alternatives beyond those represented in finite forecast ensembles.

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Chapter 1

Introduction

TODO: state the finding directly here (and abstract)

Machine-learning (ML) models have reached state-of-the-art skill in weather forecasting, rivaling traditional physics-based numerical models at a fraction of their computational cost [15, 3].

An increasingly important class of these models is formed by generative probabilistic weather models [21, 6]. Rather than producing a single deterministic forecast, these models are trained to approximate the conditional distribution of future atmospheric states. Sampling from the learned distribution enables direct generation of ensemble forecasts and thereby an explicit representation of forecast uncertainty. This probabilistic formulation also mitigates the smoothing effects often associated with deterministic models and improves probabilistic forecast skill, measured for instance by the Continuous Ranked Probability Score (CRPS) [10, 21].

Beyond their use as probabilistic samplers, generative models can also be actively controlled during generation. This is standard in image generation, where applications such as text-to-image synthesis, editing, and inpainting rely on *guidance* to steer a pretrained generative model toward a desired outcome [7, 12, 20]. Generative weather models, by contrast, are still predominantly used in their unguided form, and their potential for controlled generation remains comparatively unexplored.

A small number of recent studies have begun to investigate different forms of weather control for the generation of extreme-event scenarios, including tropical cyclones [4], precipitation extremes [27], and heatwaves [29]. These approaches shift the focus from forecasting a distribution of likely weather trajectories to generating plausible low-probability alternatives under prescribed conditions of interest. We refer to this complementary task as *scenario generation.*

This distinction is particularly relevant for extreme-weather storylines. Extreme events account for a disproportionate share of weather-related impacts [30], yet their rarity means that both historical observations and finite forecast ensembles provide only a limited sample of possible realizations. Considering plausible low-probability alternatives beyond those represented in historical observations and finite forecast ensembles is therefore essential for assessing extreme-weather risk [26, 24].

We are therefore interested in exploiting the ability of generative weather models to produce coherent weather trajectories while deliberately steering them toward prescribed extreme events. Transferring guidance methods from image generation to this setting is nonetheless not straightforward. First, while image-generation targets can often be expressed directly through a text prompt, class label, or reference image, weather scenarios require the conditioning objective to be explicitly constructed. Second, while image plausibility can often be assessed from visual coherence, weather plausibility is inherently multivariate and spatiotemporal: guided trajectories must remain coherent across space and pressure levels, preserve meaningful relationships between meteorological variables, and evolve realistically over time.

These considerations lead to the central research question of this thesis:

> *Can we steer generative probabilistic weather models to produce extreme weather trajectories that satisfy a prescribed target while remaining meteorologically plausible?*

We address this question by separating two problems that are often intertwined in guided generation: *what* scenario should be generated, and *how* the generative model should be steered to produce it. We refer to these two stages as *Specify* and *Guide*.

The first stage defines the desired scenario independently of the guidance mechanism. In our setting, this amounts to specifying a target variable, a spatial region, and a target trajectory describing how the corresponding regional quantity should evolve over forecast time. The second stage then determines how the generative dynamics must be modified to realize the prescribed target.

Within this framework, target realization becomes a guidance-strength selection problem. In contrast to common local guidance formulations, in which the magnitude of the intervention is prescribed and the resulting change in the generated sample is then observed, we specify the desired outcome first. We prescribe a relative guidance profile over flow time and adapt its overall gain separately at each weather step to reach the specified target.

We investigate this approach using ArchesWeather [6], a medium-range probabilistic weather model based on flow matching. As an experimental case study, we focus on regional surface-temperature heatwaves and specify the target through a prescribed temperature trajectory over a selected spatial region. The underlying formulation is nevertheless not restricted to temperature or extreme weather and can in principle be applied to other differentiable functionals of the forecast state and other suitably specified trajectory targets.

The main contributions of this thesis are the following.

1. We formulate the specification of a target weather scenario through a spatial mask and a scalar target trajectory, separating the desired outcome from the guidance mechanism used to realize it.
2. We develop a target-driven gradient-based guidance method that separates the overall guidance gain from its distribution over generative flow time, prescribes the relative flow-time profile, and adapts the gain at each weather step to realize the specified target.
3. We systematically investigate how the placement of guidance over flow time affects the resulting scenarios and use the selected configuration to generate and evaluate extreme-weather storylines.

The remainder of the thesis is organized as follows. Chapter 2 introduces the technical background on flow matching and guidance, reviews existing approaches to controlled weather generation, and describes ArchesWeather. Chapter 3 develops the specification of target weather scenarios, while Chapter 4 presents the guidance method used to realize them. **??** introduces the criteria used to assess the resulting trajectories, and Chapter 7 presents the experimental results, which are discussed in Chapter 8. Finally, Chapter 9 summarizes the main findings and outlines directions for future work.

Chapter 2

Background

TODO: correct equation spacing and remove equation numbering from unreferenced, or ones that do not merit

This chapter establishes the theoretical and methodological foundations for the guidance approach developed in this thesis.

We first introduce flow matching, the generative framework underlying ArchesWeather, and describe how it enables sampling from a learned data distribution. Building on this foundation, we formulate guidance from a posterior-sampling perspective, translate it to the flow-matching setting, and discuss different strategies for controlling its strength as well as alternative control-based formulations.

We then review recent approaches to the controlled generation of weather scenarios and position our work within this emerging literature. Finally, we introduce ArchesWeather, the probabilistic weather model used in this thesis, and describe the components relevant to our method, including its deterministic forecast, residual flow model, and sampling procedure.

2.1 Flow matching

The presentation in this section closely follows the exposition and notation developed by Holderrieth and Erives [13].

Generative modeling frames the task of generating objects $x_1 \in \mathbb{R}^d$ as sampling from a data distribution p_1. Flow matching [17] models p_1 as the endpoint of a continuous probability path

$$(p_t)_{t\in[0,1]},$$

which transports a tractable source distribution p_0, typically a standard Gaussian, to the data distribution p_1.

At flow time t, we denote the current state along this path by z_t, with

$$z_0 \sim p_0, \qquad z_t \sim p_t, \qquad z_1 = x_1 \sim p_1.$$

Throughout this work, we refer to z_0 as the *noise*, intermediate states z_t, $t \in (0,1)$, as *noisy states*, and the endpoint $z_1 = x_1$ as the *clean state*.

The evolution of z_t is governed by a time-dependent velocity field u_t, defining the ordinary differential equation

$$\frac{\mathrm{d}}{\mathrm{d}t} z_t = u_t(z_t). \tag{2.1}$$

Starting from $z_0 \sim p_0$ and integrating this ODE from $t = 0$ to $t = 1$ therefore transforms samples from the source distribution into samples from the data distribution.

2.1.1 Conditional probability paths

Directly specifying the marginal probability path p_t and its corresponding velocity field u_t is generally intractable. Flow matching circumvents this problem by defining a tractable *conditional probability path*

$$p_t(\cdot \mid x_1)$$

for each data sample $x_1 \sim p_1$.

A common choice is the Gaussian conditional probability path

$$p_t(\cdot \mid x_1) = \mathcal{N}\left(\alpha_t x_1,\ \beta_t^2 I\right), \tag{2.2}$$

where the schedules $\alpha_t, \beta_t \in [0,1]$ interpolate between noise and data according to

$$\alpha_0 = 0, \qquad \beta_0 = 1, \qquad \alpha_1 = 1, \qquad \beta_1 = 0.$$

Consequently,

$$p_0(\cdot \mid x_1) = \mathcal{N}(0, I), \tag{2.3}$$

$$p_1(\cdot \mid x_1) = \delta_{x_1}. \tag{2.4}$$

Sampling from the conditional path is equivalent to interpolating a data sample $x_1 \sim p_1$ with independent Gaussian noise $\varepsilon \sim \mathcal{N}(0, I)$:

$$z_t = \alpha_t x_1 + \beta_t \varepsilon \sim p_t(\cdot \mid x_1). \tag{2.5}$$

Marginalizing over the data distribution gives the corresponding marginal probability path

$$p_t(z_t) = \int p_t(z_t \mid x_1)\, p_1(x_1)\, \mathrm{d}x_1. \tag{2.6}$$

2.1.2 Conditional and marginal velocity fields

Each conditional probability path $p_t(\cdot \mid x_1)$ is generated by a corresponding conditional velocity field $u_t(z_t \mid x_1)$.
For the Gaussian path in Equation (2.2), differentiating the interpolation in Equation (2.5) gives

$$u_t(z_t \mid x_1) = \dot{\alpha}_t x_1 + \dot{\beta}_t \varepsilon. \tag{2.7}$$

Substituting

$$\varepsilon = \frac{z_t - \alpha_t x_1}{\beta_t}$$

yields

$$u_t(z_t \mid x_1) = \frac{\dot{\beta}_t}{\beta_t} z_t + \left(\dot{\alpha}_t - \frac{\dot{\beta}_t}{\beta_t} \alpha_t \right) x_1. \tag{2.8}$$

The marginal velocity field generating p_t can then be expressed as the conditional velocity averaged over all clean samples x_1 compatible with the current noisy state z_t:

$$u_t(z_t) = \mathbb{E}\left[u_t(z_t \mid x_1) \mid z_t \right]. \tag{2.9}$$

While this marginal velocity is generally not available in closed form, the conditional velocity in Equation (2.8) is tractable during training, since both x_1 and the sampled noise ε are known.

2.1.3 Flow-matching objective

Flow matching exploits the fact that a neural network can be trained on the tractable conditional velocity field while learning the corresponding marginal velocity field [17]. Writing $u_\theta(z_t, t)$ for the learned velocity field, the conditional flow-matching objective is

$$\mathcal{L}_{\mathrm{FM}}(\theta) = \mathbb{E}_{\substack{t\sim\mathcal{U}[0,1],\\ x_1\sim p_1,\\ z_t\sim p_t(\cdot|x_1)}} \left[\|u_\theta(z_t, t) - u_t(z_t \mid x_1)\|_2^2\right]. \tag{2.10}$$

This conditional objective has the same gradient with respect to the model parameters as the corresponding, but intractable, marginal flow-matching objective [17].

After training, $u_\theta(z_t, t)$ approximates the marginal velocity field $u_t(z_t)$. Generation then consists of sampling

$$z_0 \sim p_0$$

and numerically integrating

$$\frac{\mathrm{d}}{\mathrm{d}t} z_t = u_\theta(z_t, t) \tag{2.11}$$

from $t = 0$ to $t = 1$, such that the resulting endpoint z_1 approximately follows p_1.

2.1.4 Linear probability path

A particularly simple choice is the linear probability path

$$z_t = tx_1 + (1 - t)\varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I), \tag{2.12}$$

corresponding to

$$\alpha_t = t, \qquad \beta_t = 1 - t.$$

Its conditional velocity reduces to the constant displacement

$$u_t(z_t \mid x_1) = x_1 - \varepsilon. \tag{2.13}$$

Training therefore amounts to least-squares regression onto the displacement between the source noise and the clean sample.

Under the conditional probability path used to construct the marginal flow, an intermediate noisy state z_t is generally compatible with multiple clean samples x_1. Their conditional distribution is

$$p_t(x_1 \mid z_t).$$

This should be distinguished from the deterministic coupling induced by the learned flow ODE. Once a state z_t is fixed during sampling, integrating the velocity field from t to 1 determines a unique endpoint. The distribution $p_t(x_1 \mid z_t)$ instead refers to the coupling induced by the conditional probability path in Equation (2.2).

In practice, a generative model can provide a tractable clean estimate

$$\hat{x}_t := \hat{x}_\theta(z_t, t), \tag{2.14}$$

from the current noisy state. This point estimate should therefore not be confused with the deterministic endpoint obtained by continuing the learned flow from z_t. Rather, it provides a representative clean state associated with the conditional distribution $p_t(x_1 \mid z_t)$. This distinction becomes important when introducing guidance in the following section.

2.2 Guidance

Without additional guidance, a generative model samples from its base distribution $p_1(x_1)$. In many applications, however, we are interested in generating samples that satisfy an additional desired condition y. From a posterior-sampling perspective, guidance aims to modify the sampling process toward the corresponding conditional distribution

$$p_1(x_1 \mid y).$$

We first formulate guidance from this posterior-sampling perspective and translate the resulting score modification to the flow-matching velocity field. We then introduce a general guidance-strength schedule and review different strategies for choosing its magnitude over the generative trajectory. Finally, we discuss a complementary control perspective, in which guided generation is formulated as an optimization problem over variables that influence the complete generative trajectory.

2.2.1 Guidance as posterior sampling

To introduce guidance during the generative trajectory, we consider the distribution of noisy states conditioned on y at an arbitrary flow time t. By Bayes' rule,

$$p_t(z_t \mid y) \propto p_t(z_t)\, p_t(y \mid z_t). \tag{2.15}$$

Taking the gradient with respect to the noisy state gives the conditional score

$$\nabla_{z_t} \log p_t(z_t \mid y) = \nabla_{z_t} \log p_t(z_t) + \nabla_{z_t} \log p_t(y \mid z_t). \tag{2.16}$$

The first term corresponds to the unguided generative process, while the second introduces the desired condition y. Guidance can therefore be interpreted as modifying the original generation process through the likelihood gradient

$$\nabla_{z_t} \log p_t(y \mid z_t).$$

In general, however, the intermediate likelihood $p_t(y \mid z_t)$ is not directly available. We therefore marginalize over the clean sample x_1 and assume that the condition y depends on the generative process only through x_1:

$$\begin{aligned}
p_t(y \mid z_t) &= \int p_t(y \mid x_1, z_t)\, p_t(x_1 \mid z_t)\, \mathrm{d}x_1 && (2.17)\\
&= \int p(y \mid x_1)\, p_t(x_1 \mid z_t)\, \mathrm{d}x_1, \qquad y \perp z_t \mid x_1 && (2.18)\\
&= \mathbb{E}\left[p(y \mid x_1) \mid z_t\right] && (2.19)\\
&\approx \int p(y \mid x_1)\, \delta_{\hat{x}_t}(x_1)\, \mathrm{d}x_1, \qquad p_t(x_1 \mid z_t) \approx \delta_{\hat{x}_t}(x_1) && (2.20)\\
&= p(y \mid \hat{x}_t). && (2.21)
\end{aligned}$$

Here, $\hat{x}_t = \hat{x}_\theta(z_t, t)$ denotes the clean estimate obtained from the current noisy state. The approximation replaces the full conditional distribution $p_t(x_1 \mid z_t)$ by a point mass at $\hat{x}_t$. This clean-estimate approximation also underlies Diffusion Posterior Sampling [5].

The likelihood $p(y \mid x_1)$ can in principle be modeled in different ways. A convenient choice for gradient-based guidance is to define it through a differentiable loss $\mathcal{L}$:

$$p(y \mid x_1) \propto \exp\left(-\mathcal{L}(x_1, y)\right). \tag{2.22}$$

It follows that

$$\nabla_{z_t} \log p(y \mid \hat{x}_t) = -\nabla_{z_t} \mathcal{L}(\hat{x}_t, y), \tag{2.23}$$

where the gradient is propagated through the clean estimate $\hat{x}_t = \hat{x}_\theta(z_t, t)$.

2.2.2 From scores to flow-matching velocities

The previous derivation expresses guidance in terms of scores. To apply it to flow matching, we require the corresponding modification of the velocity field.

For the Gaussian probability path introduced in Equation (2.2), the marginal velocity field and score are related by [13]

$$u_t(z_t) = a_t \nabla_{z_t} \log p_t(z_t) + b_t z_t, \tag{2.24}$$

where

$$a_t := \beta_t^2 \left(\frac{\dot{\alpha}_t}{\alpha_t} - \frac{\dot{\beta}_t}{\beta_t} \right), \tag{2.25}$$

$$b_t := \frac{\dot{\alpha}_t}{\alpha_t}. \tag{2.26}$$

Applying the same relation to the conditional distribution gives

$$u_t(z_t \mid y) = a_t \nabla_{z_t} \log p_t(z_t \mid y) + b_t z_t. \tag{2.27}$$

Substituting the conditional-score decomposition yields

$$u_t(z_t \mid y) = a_t \left[\nabla_{z_t} \log p_t(z_t) + \nabla_{z_t} \log p_t(y \mid z_t)\right] + b_t z_t \tag{2.28}$$

$$= u_t(z_t) + a_t \nabla_{z_t} \log p_t(y \mid z_t). \tag{2.29}$$

Using the clean-estimate approximation derived above, we obtain

$$u_t(z_t \mid y) \approx u_t(z_t) + a_t \nabla_{z_t} \log p(y \mid \hat{x}_t). \tag{2.30}$$

Under the loss-based likelihood in Equation (2.22), this can equivalently be written as

$$u_t(z_t \mid y) \approx u_t(z_t) - a_t \nabla_{z_t} \mathcal{L}(\hat{x}_t, y). \tag{2.31}$$

Thus, the posterior-sampling formulation provides a principled local guidance direction: evaluate a loss on the clean estimate $\hat{x}_t$ and differentiate it with respect to the current noisy state z_t. The corresponding magnitude is determined by a_t under the posterior-sampling interpretation, but can more generally be treated as an algorithmic choice, as discussed next.

2.2.3 Guidance-strength schedules

The coefficient a_t in Equation (2.31) is determined by the underlying probability path. Using this coefficient preserves the posterior-sampling interpretation, up to the clean-estimate approximation introduced above.

In practice, guidance methods often rescale this contribution to obtain stronger or weaker control over the generated samples. We therefore replace a_t by a general guidance-strength schedule

$$\lambda_t := w\, \gamma_t\, \nu_t, \tag{2.32}$$

where

- $w \geq 0$ controls the overall guidance gain;
- $\gamma_t \geq 0$ controls how guidance varies along the flow;
- $\nu_t > 0$ is an optional algorithm-specific normalization factor.

We then define the *guided velocity field* as

$$\tilde{u}_t(z_t \mid y) := u_t(z_t) + \lambda_t \nabla_{z_t} \log p(y \mid \hat{x}_t) \tag{2.33}$$

$$= u_t(z_t) - \lambda_t \nabla_{z_t} \mathcal{L}(\hat{x}_t, y). \tag{2.34}$$

We refer to $u_t(z_t)$ as the *unguided velocity field.*

Setting

$$\lambda_t = a_t$$

recovers the approximate conditional velocity field in Equation (2.30). For a general choice of λ_t, however, the resulting dynamics no longer necessarily correspond to sampling from $p_1(x_1 \mid y)$. Instead, the guidance strength becomes an algorithmic parameter controlling the trade-off between the original generative dynamics and the imposed condition.

The choice of λ_t varies substantially across applications. The simplest approach is to use a constant guidance strength throughout sampling,

$$\lambda_t = w, \tag{2.35}$$

where the scalar $w > 0$ acts as a constant guidance gain and is treated as a tunable hyperparameter. Fixed guidance gains are widely used in classifier-free guidance; for instance, Esser et al. [8] evaluate rectified-flow models using several values of w. A similar strategy is used for weather guidance by Ueyama, Kawamoto, and Kera [27], who tune a single guidance gain throughout the GenCast diffusion process, omitting guidance only at the final noise level. Loss-Guided Diffusion [25] similarly tunes an overall task-level gain, while additionally introducing noise-dependent normalization to control the magnitude of the resulting guidance gradient.

Other approaches prescribe the temporal variation of the guidance strength using the noise schedule of the pretrained generative model. In the original classifier-guidance formulation [7], and subsequently in Universal Guidance [1], the guidance strength is scaled according to

$$\lambda_t = w\sqrt{1 - \bar{\alpha}_t^{\text{diff}}}, \tag{2.36}$$

where $\bar{\alpha}_t^{\text{diff}}$ denotes the cumulative signal coefficient of the pretrained diffusion process. Consequently, $\sqrt{1 - \bar{\alpha}_t^{\text{diff}}}$ is the standard deviation of the noise present at diffusion step t. The overall gain w is again tuned for the particular guidance objective, while the temporal shape of the schedule is inherited directly from the pretrained model's noise schedule rather than chosen independently.

Finally, the guidance strength can be adapted dynamically using quantities observed during sampling. Climate in a Bottle [4], for example, normalizes its classifier-guidance contribution relative to the magnitude of the unguided generative update. Schematically, in the notation used here, this corresponds to

$$\lambda_t \propto w \frac{\|u_t(z_t)\|_2}{\|\nabla_{z_t}\mathcal{L}(\hat{x}_t, y)\|_2}, \tag{2.37}$$

such that the guidance vector remains a fixed fraction of the unguided generative update. From an optimization perspective, Shen et al. [23] similarly interpret guidance strength as a step size and propose a Polyak-style adaptive rule. Their update scales the guidance gradient using both its own norm and the magnitude of the unguided model prediction, adapting the effective guidance strength at each sampling step rather than prescribing it beforehand.

These examples illustrate that, even within the same local guidance construction, the guidance-strength schedule is not uniquely determined in practice. Existing approaches range from using a constant guidance gain w, to combining w with a flow-time profile inherited from the pretrained model, to adapting the effective guidance strength dynamically during sampling.

2.2.4 Guidance as control

A complementary perspective formulates guided generation as a control problem over the generative trajectory. Rather than constructing a local approximation of a conditional score, the objective is to modify variables that influence the trajectory such that its endpoint satisfies a desired condition.

For consistency, we express the following methods using the notation introduced in this chapter rather than that of the original papers. We denote the generative state by z_t, the frozen pretrained velocity field by $u_t(z_t)$, and the terminal guidance objective by $\mathcal{L}(z_1, y)$.

FlowGrad [18] introduces an additive control c_t into the pretrained dynamics,

$$\frac{\mathrm{d}}{\mathrm{d}t} z_t = u_t(z_t) + c_t, \tag{2.38}$$

and optimizes the complete control trajectory according to

$$\min_{\{c_t\}} \ \mathcal{L}(z_1, y) + \kappa \int_0^1 \|c_t\|_2^2 \,\mathrm{d}t, \tag{2.39}$$

where c_t determines how the pretrained dynamics are modified at flow time t, and $\kappa > 0$ penalizes large deviations from the original flow. Since the terminal state z_1 depends on all preceding controls, their gradients are obtained by backpropagating the terminal objective through the ODE trajectory.

D-Flow [2] instead leaves the velocity field unchanged and optimizes only the source noise:

$$\min_{z_0} \mathcal{L}\big(z_1(z_0), y\big), \qquad \frac{\mathrm{d}}{\mathrm{d}t} z_t = u_t(z_t). \tag{2.40}$$

Here, $z_1(z_0)$ emphasizes that the generated endpoint is determined by the initial noise through integration of the frozen flow. The terminal gradient is therefore propagated through the complete generative trajectory back to z_0.

OC-Flow [28] places such approaches within a general optimal-control framework. Using the same notation, its Euclidean formulation can be viewed as optimizing a control trajectory

$$\min_{\{c_t\}} \mathcal{L}(z_1, y) + \int_0^1 \mathcal{C}(c_t)\, \mathrm{d}t, \qquad \frac{\mathrm{d}}{\mathrm{d}t} z_t = u_t(z_t) + c_t, \tag{2.41}$$

where $\mathcal{C}$ is a running cost that penalizes the applied control. OC-Flow derives the control updates using optimal-control theory and shows that existing backpropagation-through-ODE approaches can be interpreted within this framework.

The posterior-sampling and control perspectives therefore provide two complementary views of guided generation. Posterior-style methods construct local corrections to the generative dynamics from a conditioning objective, while control-based methods formulate guidance as an optimization problem over variables that influence the complete generative trajectory.

These perspectives provide the conceptual basis for the weather-specific approaches to controlled generation reviewed next.

2.3 Guided generation of weather scenarios

Having introduced the posterior-sampling and control perspectives on guidance, we now turn to their application in weather generation. Guided generation in weather and climate models remains comparatively recent, but existing work already spans several settings, from controlling individual atmospheric states to modifying complete weather trajectories. The approaches differ primarily in where control enters the generation process and how the desired event is specified.

For multi-step weather trajectories, we use n to index weather time, while t denotes the internal diffusion or flow time of the generative process. Sampling-time conditioning has also been explored beyond extreme-event generation, for example by incorporating external forecast information during diffusion sampling [14]. Here, we focus specifically on methods aimed at controlling rare or extreme weather outcomes.

State-level guidance. At the most local level, guidance can be used to control the properties of a single generated atmospheric state. A representative example is Climate in a Bottle (cBottle) [4], which uses classifier guidance for the on-demand generation of extreme atmospheric states. Unlike autoregressive weather models, cBottle generates independent multivariate states conditioned on slowly varying quantities such as sea-surface temperature and solar position.

To generate tropical cyclones at user-specified locations, the authors train a classifier on historical cyclone observations and use its gradient during diffusion sampling. Expressed schematically in the notation introduced above, the guided dynamics take the form

$$\tilde{u}_t(z_t \mid y) = u_t(z_t) - \lambda_t \nabla_{z_t} \mathcal{L}_{\mathrm{TC}}(z_t, y), \tag{2.42}$$

where y specifies the desired tropical-cyclone locations and $\mathcal{L}_{\mathrm{TC}}$ denotes the classifier loss. The guidance strength is adaptively normalized relative to the unguided generative update, as discussed in Section 2.2.3. The resulting samples contain cyclones at the requested locations while retaining correlated structure across atmospheric variables [4].

Since cBottle generates atmospheric states independently, the method controls the occurrence and location of an event in a single state rather than its temporal evolution.

Forecast guidance. Guidance has also been applied directly within probabilistic weather forecasting. Ueyama, Kawamoto, and Kera [27] use gradient-based guidance during GenCast diffusion sampling to reduce precipitation over a prescribed target region. In contrast to approaches that perturb an atmospheric state before forecasting, the intervention is applied within the generative sampling process itself.

Using the guidance notation introduced above, the mechanism can be represented schematically as

$$\tilde{u}_t(z_t \mid y) = u_t(z_t) - \lambda \nabla_{z_t} \mathcal{L}_{\mathrm{precip}}(\hat{x}_t, y), \tag{2.43}$$

where $\mathcal{L}_{\mathrm{precip}}$ measures the precipitation-reduction objective on the current clean estimate $\hat{x}_t$. The gradient is propagated through the clean estimate to the noisy state z_t and used to modify the generative update.

The guidance strength λ is fixed during sampling and controls the magnitude of the intervention. Increasing it produces stronger precipitation reduction, but also larger deviations from the original forecast. Accordingly, the authors evaluate not only whether the intervention succeeds, but also its effects on the surrounding atmospheric state through variable-wise and vertical perturbation structures, deviations of the generative trajectory, and cross-model transferability [27].

Trajectory-level control. Control objectives can also be defined over complete weather trajectories. A particularly relevant example is Whittaker and Di Luca [29], who construct extreme heatwave storylines by exploiting the differentiability of NeuralGCM. Starting from a historical atmospheric state x_0, they optimize a small perturbation δx_0 to the initial condition whose subsequent evolution produces a more extreme heatwave:

$$\delta x_0^\star = \arg\min_{\delta x_0} \left[\mathcal{L}_{\mathrm{HW}}\big(F_{0:N}(x_0 + \delta x_0)\big) + \kappa \|\delta x_0\|_2^2\right], \tag{2.44}$$

where $F_{0:N}$ denotes the differentiable NeuralGCM rollout, $\mathcal{L}_{\mathrm{HW}}$ penalizes trajectories that fail to attain the desired heatwave intensity, and $\kappa > 0$ controls the penalty on the magnitude of the initial perturbation.

Gradients are propagated through the complete model trajectory back to the initial atmospheric state. Applied to the 2021 Pacific Northwest heatwave, the optimization produces trajectories more extreme than those found in a conventional ensemble while retaining characteristic large-scale structures of the event [29].

This construction differs from the generative-guidance approaches above. NeuralGCM is used as a differentiable dynamical model, and control is achieved by optimizing the initial atmospheric condition rather than by intervening within a generative sampling process. It therefore provides an example of trajectory-level control through initial-condition optimization.

Other approaches control trajectories without differentiating an event loss through the forecast model. He and Guo [9] use Intensity–Duration–Frequency profiles as likelihood constraints during sequential generation, preferentially retaining trajectories compatible with the desired extreme profile. Related rare-event methods instead clone or prune trajectories according to their likelihood of developing into the target event [16].

A further alternative is to incorporate the desired condition during training rather than impose it on a pretrained generator at inference time. Loer, Schillinger, and Sippel [19], for example, learn a conditional generative distribution of temperature fields given atmospheric circulation and global warming level. In the Bayesian terminology introduced above, this corresponds to learning $p(x \mid y)$ directly, rather than starting from a pretrained model of $p(x)$ and introducing the condition through $p(y \mid x)$ during sampling.

Taken together, these studies illustrate several routes to controlled weather generation: guiding individual atmospheric states, intervening during probabilistic forecast generation, optimizing initial conditions for complete trajectories, selectively sampling rare trajectories, and learning conditional generators directly.

2.4 ArchesWeather

We investigate this problem using ArchesWeather [6], a probabilistic medium-range weather forecasting model that combines a deterministic forecast with a flow-matching model of the forecast residual. Having established the general flow-matching and guidance formulations above, we now describe the components of ArchesWeather required for the development of our method and refer to the original work for architectural and training details.

2.4.1 Dataset

ArchesWeather is trained on the ERA5 reanalysis dataset [11], using one of the versions preprocessed by WeatherBench 2 [22]. The data are represented on a regular 1.5° latitude–longitude grid with 121×240 spatial points.

Each weather state

$$x_n \in \mathbb{R}^{82\times121\times240}$$

contains 82 atmospheric fields. Four are surface variables: 2 m temperature, mean sea-level pressure, and the two components (u and v) of the 10 m wind. The remaining 78 fields correspond to six upper-air variables

$$\{\text{geopotential}, \text{temperature}, \text{specific humidity}, u, v, \text{vertical velocity}\}$$

evaluated at the 13 pressure levels

$$\{50, 100, 150, 200, 250, 300, 400, 500, 600, 700, 850, 925, 1000\}\ \text{hPa}.$$

Weather states are available at 00:00, 06:00, 12:00, and 18:00 UTC. Following the WeatherBench 2 evaluation protocol, dates from January 2020 onward are reserved for evaluation.

Both the deterministic and generative components operate in normalized data space. Unless stated otherwise, model states introduced below therefore refer to their normalized representation. Quantities defined in physical units are evaluated after applying the corresponding inverse normalization. For notational simplicity, this transformation is treated as part of the respective diagnostic operator and is omitted from subsequent notation.

2.4.2 Forecasting setup

ArchesWeather generates forecast trajectories autoregressively. Following the convention introduced above, we index forecast lead time by n, which we refer to as *weather time*. At each weather step, the model produces the state x_n conditioned on the two preceding states x_{n-2} and x_{n-1}.

Rather than generating x_n directly, ArchesWeather decomposes the forecast into a deterministic component and a stochastic residual component. The deterministic model first produces a base forecast

$$\hat{x}_n^{\mathrm{det}}.$$

The corresponding ground-truth residual is defined as

$$r_n := x_n - \hat{x}_n^{\mathrm{det}}. \tag{2.45}$$

A conditional flow-matching model is then trained to generate this residual. At inference time, a sampled residual $\hat{r}_n$ is combined with the deterministic prediction as

$$\hat{x}_n = \hat{x}_n^{\mathrm{det}} + \hat{r}_n. \tag{2.46}$$

For a fixed conditioning context, sampling different residuals while keeping the deterministic forecast fixed therefore produces different ensemble members at the current weather step.

2.4.3 Residual flow model

Generating each stochastic residual requires a complete flow-matching trajectory. ArchesWeather therefore involves two nested notions of time: the weather-time index n, which advances the autoregressive forecast, and the flow time t, which describes the generation of a residual within each weather step.

Following the notation introduced in the previous sections, the clean endpoint of the flow is now the forecast residual r_n, while $z_{n,t}$ denotes its noisy state at flow time t during the generation of weather state x_n.

ArchesWeather parameterizes the Gaussian probability path in terms of a noise level s_t, which decreases linearly from 1 at the beginning of the flow to 0 at its endpoint. The noisy residual is constructed as

$$z_{n,t} = (1 - s_t)\, r_n + s_t\, \varepsilon, \qquad \varepsilon \sim \mathcal{N}(0, I). \tag{2.47}$$

This corresponds directly to the Gaussian probability path introduced in Equation (2.2), with

$$\alpha_t = 1 - s_t, \qquad \beta_t = s_t. \tag{2.48}$$

The flow therefore starts from Gaussian noise and progressively approaches the clean residual r_n.

Unlike the velocity-field parameterization introduced in the generic formulation above, the ArchesWeather residual model is trained to predict the clean residual from the noisy state. We denote this estimate by

$$\hat{r}_{n,t} := r_\theta(z_{n,t}, t, c_n), \tag{2.49}$$

where

$$c_n = \left(x_{n-2}, x_{n-1}, \hat{x}_n^{\mathrm{det}}\right)$$

collects the conditioning information provided to the residual model.

The model is trained using the clean-prediction objective

$$\mathcal{L}_{\mathrm{Arches}}(\theta) = \mathbb{E}_{t, r_n, \varepsilon} \left[\| r_\theta(z_{n,t}, t, c_n) - r_n \|_2^2 \right]. \tag{2.50}$$

In the notation of the previous sections, $\hat{r}_{n,t}$ thus plays the role of the clean estimate $\hat{x}_t$. Since the flow generates a residual rather than the complete weather state, the corresponding clean weather estimate is

$$\hat{x}_{n,t} := \hat{x}_n^{\mathrm{det}} + \hat{r}_{n,t}. \tag{2.51}$$

2.4.4 Sampling

Although the network predicts the clean residual, sampling requires the corresponding velocity field. From Equation (2.47), the noise associated with the clean estimate $\hat{r}_{n,t}$ is

$$\hat{\varepsilon}_{n,t} = \frac{z_{n,t} - (1 - s_t)\hat{r}_{n,t}}{s_t}. \tag{2.52}$$

For the linear path

$$\alpha_t = 1 - s_t, \qquad \beta_t = s_t,$$

the velocity is the displacement between the clean residual estimate and the corresponding noise estimate:

$$\begin{aligned} u_{n,t} &:= u_\theta(z_{n,t}, t, c_n) && (2.53)\\ &= \hat{r}_{n,t} - \hat{\varepsilon}_{n,t} && (2.54)\\ &= \hat{r}_{n,t} - \frac{z_{n,t} - (1 - s_t)\hat{r}_{n,t}}{s_t} && (2.55)\\ &= \frac{\hat{r}_{n,t} - z_{n,t}}{s_t}. && (2.56) \end{aligned}$$

ArchesWeather discretizes the residual flow into $T = 25$ sampling steps, indexed by $t = 0, \dots, T-1$. At sampling step t, the model evaluates the current residual state $z_{n,t}$ and advances it to $z_{n,t+1}$.

Starting from Gaussian noise

$$z_{n,0} \sim \mathcal{N}(0, I),$$

the residual is generated through Euler integration,

$$z_{n,t+1} = z_{n,t} + h_t\, u_{n,t}, \qquad h_t = s_t - s_{t+1}, \qquad t = 0, \dots, T-1. \tag{2.57}$$

Since the noise level decreases from $s_0 = 1$ to $s_T = 0$, the residual states progressively evolve from Gaussian noise toward a clean residual.

After the T sampling steps, the endpoint $z_{n,T}$ is taken as the sampled residual,

$$\hat{r}_n := z_{n,T}, \tag{2.58}$$

which is added to the deterministic forecast to obtain

$$\hat{x}_n = \hat{x}_n^{\text{det}} + \hat{r}_n = \hat{x}_n^{\text{det}} + z_{n,T}. \tag{2.59}$$

Figure 2.1 summarizes the computations performed within a single weather step. This structure will be central to the implementation and evaluation of our guidance method.

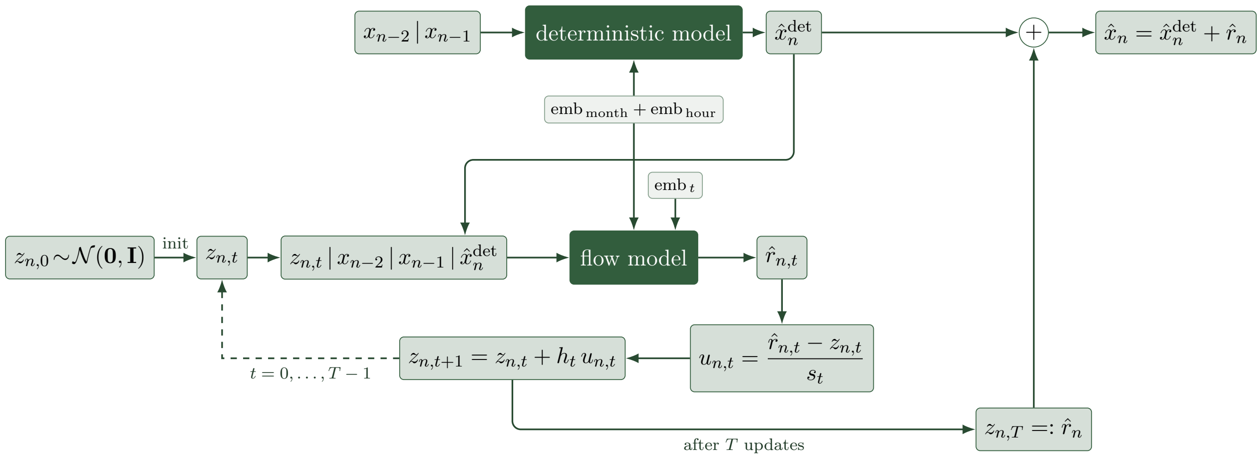


Figure 2.1 – Computational graph of one ArchesWeather forecast step. The deterministic model produces the base forecast $\hat{x}_n^{\text{det}}$, while the flow model generates the residual through T sampling steps before both components are combined.

The nested residual-flow structure of ArchesWeather provides the concrete generative process on which our guidance method will operate. Before developing that method in Chapter 4, we first specify the weather trajectories toward which the model should be steered in Chapter 3.

Chapter 3

Specifying events of interest

The first stage of our approach addresses *what* weather scenario should be generated. We specify an event through two components: a mask π, which selects the atmospheric variable and spatial region of interest, and a relative intensity trajectory $\rho_{1:H}$, which describes how strongly the targeted quantity should deviate from an unguided reference over the guided forecast horizon.

Throughout this thesis, we focus on regional high-temperature scenarios. Accordingly, π selects surface temperature over a prescribed region, and $\rho_{1:H}$ controls the requested increase of its regional mean. The underlying formulation is not restricted to temperature: the regional mean used here could be replaced by another differentiable functional of the forecast state.

Since ArchesWeather is a generative model, the same initial atmospheric context can produce an ensemble of different forecast trajectories. We therefore avoid assigning every ensemble member the same absolute target. Instead, we first generate an unguided reference ensemble and apply a common relative change ρ_n to each member's own reference value. This yields member-specific absolute targets while retaining the differences present across the reference ensemble.

The relative change ρ_n can be interpreted through the corresponding change in the ensemble-mean targeted quantity. In the experiments considered here, it serves as a controlled intensity parameter rather than as a climatological definition of extremeness. Numerical choices of its magnitude are introduced together with the experimental setup.

We first define the target variable and spatial region through the mask π. We then construct the relative intensity trajectory and derive the member-specific targets used by the guidance algorithm. Finally, we summarize the complete specification pipeline from the unguided reference ensemble to the resulting target trajectories.

3.1 Target variable and region

We specify the atmospheric quantity of interest through a mask π over the weather state

$$x_n \in \mathbb{R}^{V \times I \times J},$$

where V indexes atmospheric fields and I and J index latitude and longitude. The corresponding spatial coordinates are denoted by

$$(\varphi_i, \psi_j), \qquad i = 1, \dots, I, \quad j = 1, \dots, J.$$

Let $v^\star$ denote the target variable and let

$$k_{ij} \in \{0, 1\}$$

indicate whether grid cell (i, j) lies inside the selected target region. In this thesis, the spatial support is defined by a rectangular bounding box on the latitude–longitude grid.

The mask is non-zero only for the selected variable and within the selected region. To obtain a regional average that accounts for differences in the physical area represented by each grid cell, we weight each included latitude row according to

$$a_i = \int_{\beta_i^-}^{\beta_i^+} \cos\varphi \, \mathrm{d}\varphi, \tag{3.1}$$

where $[\beta_i^-, \beta_i^+]$ denotes the latitude extent of row i. Because the longitudinal grid spacing is constant, its contribution cancels under normalization.

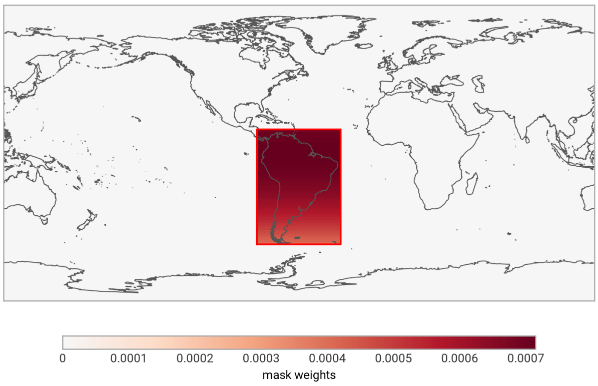
We therefore define the normalized mask as

$$\pi_{vij} = \frac{\mathbf{1}[v = v^\star] \, k_{ij} \, a_i}{\sum_{i',j'} k_{i'j'} \, a_{i'}}, \tag{3.2}$$

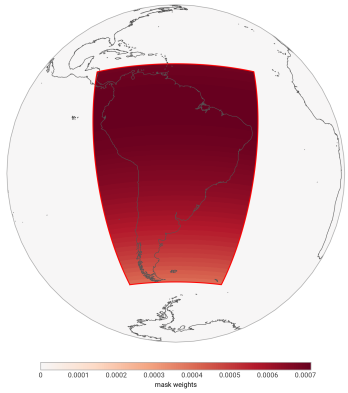
such that

$$\sum_{v,i,j} \pi_{vij} = 1.$$

The corresponding masked-average operator is



(a) Latitude–longitude representation.



(b) Spherical representation.

Figure 3.1 – Example of the spatial target mask. The rectangular support determines which grid cells are included, while the latitude-dependent weights account for differences in physical grid-cell area.

$$\mathcal{M}_\pi(x_n) := \sum_{v,i,j} \pi_{vij}\, x_{n,vij}. \tag{3.3}$$

Following the convention introduced in the ArchesWeather dataset description, the inverse normalization required to evaluate this diagnostic in physical units is implicit in $\mathcal{M}_\pi$.

For the experiments in this thesis, $v^\star$ is surface temperature, such that $\mathcal{M}_\pi(x_n)$ represents the area-weighted mean surface temperature within the selected target region.

Area weighting prevents the regional mean from being biased toward high-latitude grid cells, which represent a smaller physical area on the sphere. This follows the latitude-weighting convention used in WeatherBench 2 [22].

Figure 3.1 illustrates the resulting mask for an example target region over South America. Its support is rectangular on the latitude–longitude grid, while the normalized weights vary with latitude according to the physical area represented by each grid cell.

The binary spatial support treats all locations within the selected region equally apart from the correction for physical grid-cell area. Normalizing π to sum to one ensures that $\mathcal{M}_\pi$ remains an average and is therefore directly comparable across regions of different sizes.

Although we use $\mathcal{M}_\pi$ to compute a regional mean, the subsequent guidance formulation does not fundamentally depend on this particular choice. It can in principle be replaced by another differentiable functional of the forecast state.

3.2 Target trajectory

Once the target variable and region have been specified through π, the remaining task is to define how strongly the targeted quantity should deviate from an unguided reference over weather time.

Let M denote the number of ensemble members, N the full experimental forecast horizon, and $H \leq N$ the final guided weather step. Guidance is applied for $n = 1, \ldots, H$, while the forecast is continued without guidance for $n > H$. This allows us to examine how the generated scenario evolves after the intervention is removed.

3.2.1 Reference ensemble and member-specific targets

Before generating a guided ensemble, we first generate a complete unguided reference ensemble

$$\left\{x_{n,m}^{\text{ung}}\right\}_{m=1,\ldots,M}$$

over the experimental horizon, with $n = 1, \ldots, N$.

All members start from the same initial atmospheric context but differ through their stochastic residual realizations. For each member and weather step, the initial residual-noise realization used in the reference rollout is retained and later reused when generating the corresponding guided member. The stochastic realizations are therefore paired, although the autoregressive conditioning states generally differ once guidance has altered the trajectory.

For convenience, we denote the targeted quantity of reference member m at weather step n by

$$q_{n,m}^{\text{ung}} := \mathcal{M}_\pi\left(x_{n,m}^{\text{ung}}\right). \tag{3.4}$$

Rather than assigning every member the same absolute target, we prescribe a common relative change ρ_n and apply it to each member's own reference value:

$$y_{n,m}^\star := (1 + \rho_n)\, q_{n,m}^{\text{ung}}, \qquad n = 1, \ldots, H. \tag{3.5}$$

The member-specific targets therefore retain the differences present across the unguided reference ensemble. Members with different reference values are assigned correspondingly different absolute targets instead of being forced toward a common trajectory. This property concerns the construction of the target ensemble; whether the resulting guided ensemble retains the implied spread depends on the behaviour of the guidance algorithm.

3.2.2 Prescribing the relative change

The dimensionless quantity ρ_n controls the requested intensity of the scenario at weather step n. Its magnitude can be related to a more physically interpretable change in the ensemble-mean targeted quantity.

Define the unguided ensemble mean as

$$\bar{q}_n^{\text{ung}} := \frac{1}{M} \sum_{m=1}^{M} q_{n,m}^{\text{ung}}. \tag{3.6}$$

Under Equation (3.5), the mean of the member-specific targets is

$$\bar{y}_n^\star = (1 + \rho_n)\, \bar{q}_n^{\text{ung}}. \tag{3.7}$$

The corresponding requested change in the ensemble mean is therefore

$$\Delta \bar{q}_n^\star := \bar{y}_n^\star - \bar{q}_n^{\text{ung}} = \rho_n\, \bar{q}_n^{\text{ung}}. \tag{3.8}$$

Equivalently, a desired ensemble-mean change can be translated into the relative intensity parameter through

$$\rho_n = \frac{\Delta \bar{q}_n^\star}{\bar{q}_n^{\text{ung}}}. \tag{3.9}$$

For the surface-temperature experiments in this thesis, $\Delta \bar{q}_n^\star$ corresponds to a requested increase in regional mean temperature. We nevertheless use ρ_n as the parameter of the specification, since the same construction can be applied independently of the units of the targeted quantity.

In principle, the trajectory $\rho_{1:H}$ could be chosen from historical events, climatological statistics, or another application-specific definition of the desired scenario. Here, however, our aim is to study the guidance method under controlled and progressively more demanding conditions. We therefore use the simple linear trajectory

$$\rho_n := \frac{n}{H} \rho_H, \qquad n = 1, \ldots, H, \tag{3.10}$$

where ρ_H denotes the requested relative intensity at the final guided weather step.

The linear trajectory gradually increases the requested deviation rather than imposing its full magnitude from the beginning of the forecast. It is chosen for experimental simplicity and interpretability rather than as a model of the temporal evolution of real high-temperature events. Different choices of ρ_H therefore define different experimental intensity levels, whose numerical values are introduced in the experimental setup.

After weather step H, no target is imposed. The forecast is nevertheless continued until N, allowing us to assess how the guided trajectory evolves once guidance is switched off.

3.3 Pipeline overview

The complete specification procedure is summarized in Figure 3.2. Starting from an initial atmospheric context (x_{-1}, x_0), we first choose the mask π, which determines the target variable and spatial region.

For each ensemble member $m = 1, \ldots, M$, we then generate an unguided reference rollout over the full experimental horizon,

$$\left\{x_{n,m}^{\text{ung}}\right\}_{n=1}^{N}.$$

Together with the mask, this reference ensemble determines the corresponding regional quantities

$$q_{n,m}^{\text{ung}} = \mathcal{M}_{\pi}\left(x_{n,m}^{\text{ung}}\right),$$

and their ensemble mean $\bar{q}_n^{\text{ung}}$.

We next prescribe the relative intensity trajectory $\rho_{1:H}$. As discussed in Section 3.2, its magnitude can equivalently be motivated through a desired change $\Delta\bar{q}_n^{\star}$ in the ensemble-mean targeted quantity, with

$$\rho_n = \frac{\Delta\bar{q}_n^{\star}}{\bar{q}_n^{\text{ung}}}.$$

Applying the resulting relative change to each reference member yields the member-specific target trajectories

$$\left\{y_{1:H,m}^{\star}\right\}_{m=1}^{M},$$

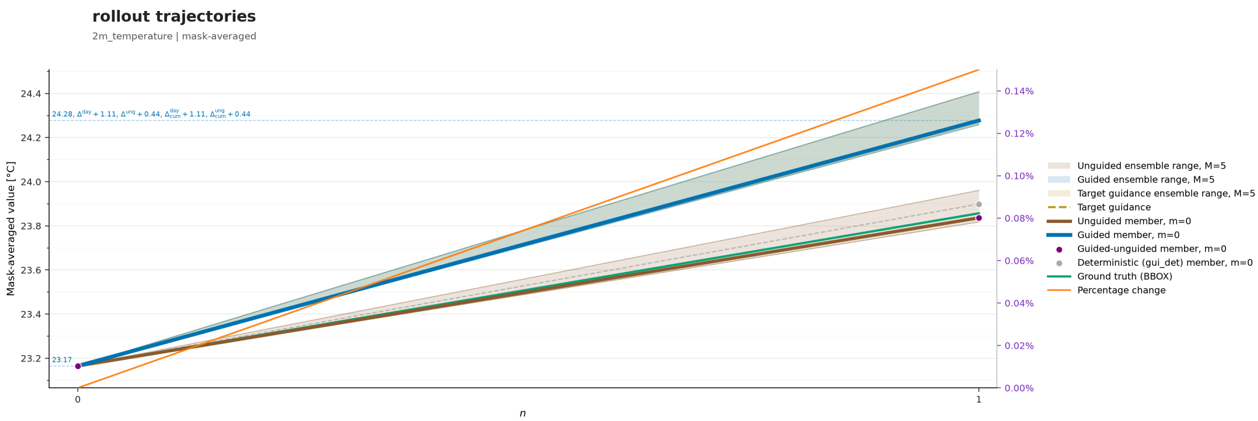


Figure 3.2 – Schematic illustration of the specification procedure. Each unguided reference member defines a baseline trajectory for the targeted regional quantity. Applying the common relative intensity trajectory $\rho_{1:H}$ produces a corresponding member-specific target trajectory over the guided horizon. No target is imposed after weather step H, while the experiment continues until step N.

where

$$y^{\star}_{n,m} = (1 + \rho_n)\, q^{\text{ung}}_{n,m}.$$

The scenario specification is therefore characterized by the target mask π and relative intensity trajectory $\rho_{1:H}$. For a given unguided reference ensemble, these induce the absolute member-specific targets

$$\left\{ y^{\star}_{1:H,m} \right\}_{m=1}^{M},$$

which are passed to the guidance stage developed in the next chapter.

Guidance is applied only for $n = 1, \ldots, H$. The resulting forecast is then continued without guidance until the experimental horizon N, allowing us to examine how the generated scenario evolves after the intervention has ended.

This completes the specification of *what* the model should generate. The next chapter addresses *how* the residual flow is modified to realize these member-specific targets.

Chapter 4

Guidance design

The previous chapter specified the scenario of interest through a target mask π and relative intensity trajectory $\rho_{1:H}$. Together with the unguided reference ensemble, these determine the member-specific target trajectories $y^{\star}_{1:H,m}$. We now address the second component of the framework: how the generative forecast is steered toward these targets.

Because guidance is applied independently to each ensemble member, we suppress the member index throughout this chapter. Accordingly, $y^{\star}_{n}$ denotes the target associated with the member currently being generated.

We formulate this problem as *target-driven guidance*. At each weather step, the discrepancy between the prescribed target and intermediate clean-state estimates provides a local direction in which to modify the generative flow. Rather than fixing the magnitude of this intervention beforehand, we prescribe how guidance is distributed over flow time and adapt its overall gain to realize the target at the end of the flow.

This separates two aspects of the guidance-strength design: a flow-time profile γ_t, which determines the relative placement of guidance within the generative trajectory, and a weather-step-specific gain w_n, which determines its overall magnitude. Once the profile has been chosen, target realization therefore reduces to adapting a single scalar gain while generating each weather state.

We first describe how guidance is incorporated into the autoregressive rollout of ArchesWeather and define the local guidance objective. We then introduce the guidance-strength design and discuss alternative ways of controlling the intervention over flow time. Finally, we formulate target realization as a scalar root-finding problem and adapt w_n independently at each guided weather step.

4.1 Target-driven guidance

ArchesWeather generates forecast trajectories autoregressively. At weather step n, the states x_{n-2} and x_{n-1} form the atmospheric context from which x_n is generated. Guidance acts within the residual flow used to generate this state by modifying its velocity field.

Once a guided state x_n^{gui} has been generated, it becomes part of the conditioning context used at subsequent weather steps. The effects of guidance at each weather step therefore propagate throughout the remaining part of the rollout.

To distinguish the different trajectories involved in this process, we introduce three weather-state objects. We denote by

$$x_n^{\mathrm{ung}} \equiv x_n^{\mathrm{ung|ung}}$$

the state of the unguided reference rollout introduced in Chapter 3. Both the current residual flow and the preceding atmospheric states are unguided. This rollout provides the reference from which the member-specific target $y_n^{\star}$ is constructed.

We denote by

$$x_n^{\mathrm{gui}} \equiv x_n^{\mathrm{gui|gui}}$$

the state obtained by applying guidance to the current residual flow while conditioning on the preceding guided states.

Finally, we define

$$x_n^{\mathrm{ung|gui}}$$

as the state obtained by running the current residual flow without guidance while retaining the same guided atmospheric context used to generate x_n^{gui}. The two states $x_n^{\mathrm{ung|gui}}$ and x_n^{gui} therefore share the same preceding weather states, deterministic forecast, and initial residual-noise realization $z_{n,0}$; they differ only in whether guidance is applied within the current residual flow. This state will later provide the zero-gain evaluation of the target-realization procedure.

At a given weather step, the same initial residual-noise realization is also used for the corresponding state of the unguided reference rollout. Thus, x_n^{ung} and x_n^{gui} remain paired through their stochastic realization, while their atmospheric conditioning contexts generally differ once guidance has altered the preceding trajectory.

The distinction between x_n^{ung} and $x_n^{\mathrm{ung|gui}}$ therefore becomes important after the guided and unguided trajectories have diverged. Their deterministic forecasts are, respectively,

$$\hat{x}_n^{\mathrm{det|ung}} = F_{\mathrm{det}}\left(x_{n-2}^{\mathrm{ung}}, x_{n-1}^{\mathrm{ung}}\right), \tag{4.1}$$

$$\hat{x}_n^{\mathrm{det|gui}} = F_{\mathrm{det}}\left(x_{n-2}^{\mathrm{gui}}, x_{n-1}^{\mathrm{gui}}\right). \tag{4.2}$$

The latter deterministic forecast is shared by $x_n^{\mathrm{ung|gui}}$ and x_n^{gui}, since their difference arises only from the guidance applied within the residual flow.

At the first generated weather state, $n = 1$, the unguided reference and unguided-under-guided-context constructions share both the initial atmospheric context (x_{-1}, x_0) and the same initial residual noise. Consequently,

$$x_1^{\mathrm{ung|gui}} = x_1^{\mathrm{ung}}. \tag{4.3}$$

For $n > 1$, this equality no longer holds in general, since the preceding guided states can differ from their unguided reference counterparts.

TODO: Insert summary figure illustrating $x_n^{ung}, x_n^{ung|gui}, x_n^{gui}, \hat{x}_n^{det|ung}, \hat{x}_n^{det|gui}, y_n^\star$.

4.1.1 Guidance objective

At weather step n, the specification procedure introduced in Chapter 3 provides the member-specific target value $y_n^\star$. During the residual flow, ArchesWeather provides the corresponding clean-state estimate $\hat{x}_{n,t}$ from the current noisy residual state $z_{n,t}$.

We define the discrepancy from the prescribed target as

$$\xi_{n,t} := \mathcal{M}_\pi\left(\hat{x}_{n,t}\right) - y_n^\star, \tag{4.4}$$

where $\mathcal{M}_\pi$ denotes the area-weighted regional mean operator defined in Chapter 3. We refer to $\xi_{n,t}$ as the *target gap* at flow step t.

We use the squared target gap as the guidance loss,

$$\mathcal{L}_{n,t} := \xi_{n,t}^2 = \left[\mathcal{M}_\pi\left(\hat{x}_{n,t}\right) - y_n^\star\right]^2. \tag{4.5}$$

Following the clean-estimate approximation derived in Equation (2.21), the objective is evaluated on the current clean-state estimate rather than directly on the unknown clean endpoint.

An alternative would be to define the objective only on the final generated state and propagate its gradient backwards through the complete residual flow. This corresponds to the control perspective discussed in Section 2.2.4. Such a formulation requires differentiating the terminal objective through the sequence of flow updates in order to determine interventions at earlier states. The local clean-estimate formulation instead obtains the guidance direction from the current model evaluation alone. We retain this posterior-style construction and use the terminal target discrepancy separately to determine the overall magnitude of guidance.

For the experiments in this thesis, $\mathcal{M}_\pi$ represents the area-weighted regional mean surface temperature introduced in Chapter 3. The guidance construction itself is more general and can in principle be applied to other differentiable functionals of the forecast state.

4.1.2 Gradient-based flow guidance

For each weather state x_n, the residual flow starts from Gaussian noise

$$z_{n,0} \sim \mathcal{N}(0, I)$$

and produces the final residual state $z_{n,T}$ through T sampling steps, indexed by $t = 0, \ldots, T-1$. At sampling step t, ArchesWeather evaluates the velocity $u_{n,t}$ and clean-state estimate from the current noisy residual state $z_{n,t}$ before advancing to $z_{n,t+1}$.

Following the loss-based guidance formulation introduced in the background chapter, we define the local guidance gradient as

$$g_{n,t} := \nabla_{z_{n,t}} \mathcal{L}_{n,t}. \tag{4.6}$$

For the squared target-gap loss in Equation (4.5), this gives

$$g_{n,t} = 2\xi_{n,t} \nabla_{z_{n,t}} \mathcal{M}_\pi \left(\hat{x}_{n,t}\right). \tag{4.7}$$

Although $\mathcal{M}_\pi$ evaluates the targeted quantity in physical units, differentiation is performed with respect to the normalized noisy residual state $z_{n,t}$. The gradient therefore implicitly propagates through the inverse normalization as well as the learned mapping from the noisy residual to the clean-state estimate. The resulting guidance direction consequently lives in the normalized residual space in which the generative flow evolves.

Moreover, although the objective depends only on the regional quantity selected by π, the gradient is taken with respect to the complete residual state. It can therefore contain non-zero components outside the explicitly targeted variable and region. Rather than directly perturbing the selected atmospheric entries, the guidance gradient describes how the current residual state should change, through the learned generative model, to modify the targeted regional quantity.

The velocity field is then modified according to

$$\tilde{u}_{n,t} = u_{n,t} - \lambda_{n,t} g_{n,t}, \tag{4.8}$$

where $\lambda_{n,t} \geq 0$ denotes the *guidance strength* at weather step n and flow step t. The corresponding Euler update becomes

$$z_{n,t+1} = z_{n,t} + h_t \tilde{u}_{n,t}, \qquad t = 0, \ldots, T-1. \tag{4.9}$$

The guidance gradient $g_{n,t}$ is therefore determined locally by the prescribed target and the current generative state. What remains to be specified is the guidance strength $\lambda_{n,t}$ with which this direction is applied over the flow.

4.2 Guidance-strength design

The general formulation in Section 2.2.3 decomposes the guidance strength into an overall gain, a flow-time profile, and an optional normalization factor. Applied independently at each weather step, this gives

$$\lambda_{n,t} = w_n \gamma_t \nu_{n,t}, \tag{4.10}$$

where w_n is the overall *guidance gain* used while generating x_n, γ_t determines the relative variation of guidance over flow time, and $\nu_{n,t}$ represents an optional additional normalization.

In our method, we apply the raw guidance gradient without additional normalization and therefore set

$$\nu_{n,t} = 1. \tag{4.11}$$

The guidance strength consequently reduces to

$$\lambda_{n,t} = w_n \gamma_t, \tag{4.12}$$

and the guided velocity in Equation (4.8) becomes

$$\tilde{u}_{n,t} = u_{n,t} - w_n \gamma_t g_{n,t}. \tag{4.13}$$

4.2.1 Final flow step

Guidance is applied during the first $T-1$ Euler updates but omitted from the final update, indexed by $t = T-1$. This convention follows from the clean-prediction parameterization of ArchesWeather.

At the beginning of the final update, the residual state is $z_{n,T-1}$ and the remaining noise level is s_{T-1}. From Equation (2.56), the corresponding velocity is

$$u_{n,T-1} = \frac{\hat{r}_{n,T-1} - z_{n,T-1}}{s_{T-1}}. \tag{4.14}$$

Since the endpoint has noise level $s_T = 0$, the final Euler step size is

$$h_{T-1} = s_{T-1} - s_T = s_{T-1}. \tag{4.15}$$

Without guidance, the final update therefore becomes

$$\begin{aligned} z_{n,T} &= z_{n,T-1} + s_{T-1} u_{n,T-1} &(4.16)\\ &= z_{n,T-1} + \hat{r}_{n,T-1} - z_{n,T-1} &(4.17)\\ &= \hat{r}_{n,T-1}. &(4.18) \end{aligned}$$

The final update thus maps the noisy residual state directly onto the current clean-residual prediction. Applying guidance at this step would perturb the resulting residual without any subsequent model evaluation in which the modified state could be processed. We therefore set

$$\gamma_{T-1} = \lambda_{n,T-1} = 0. \tag{4.19}$$

Hence, $t = T-2$ is the final guided flow step. Omitting guidance at the final noise level is also consistent with the precipitation-guidance procedure of Ueyama, Kawamoto, and Kera [27].

4.2.2 Flow-time profile

We prescribe γ_t as a non-negative profile over the guided flow steps. Since its overall scale can be absorbed into w_n, we normalize it such that

$$\max_{t=0,\ldots,T-2} \gamma_t = 1. \tag{4.20}$$

The two components of the guidance strength therefore have distinct roles: γ_t determines *where* along the generative flow guidance is applied, whereas w_n determines its overall magnitude while generating the current weather state. The profile families considered in this thesis are introduced together with the experimental design.

4.2.3 Alternative target-realization strategies

TODO: choose how much to say and whether to defer to an appendix

The decomposition above leaves several possibilities for determining the guidance strengths required to realize the prescribed target. During method development, we considered three ways of doing so.

A first approach adapts the guidance strength locally at every flow step such that the remaining target gap follows a prescribed trajectory toward zero. This directly controls the evolution of the target gap during generation, but requires specifying how quickly it should close within the internal flow.

A second possibility treats the individual guidance strengths

$$\lambda_{n,0}, \ldots, \lambda_{n,T-2}$$

as independent optimization variables and determines them jointly from the terminal target discrepancy. This provides greater flexibility over the placement of guidance but introduces a higher-dimensional optimization problem.

The formulation retained in this thesis instead fixes the relative flow-time structure through γ_t and adapts only the scalar guidance gain w_n. This reduces target realization to a one-dimensional problem while preserving explicit control over where guidance is applied during the flow. The empirical implications of this restriction are examined in the experimental analysis.

4.3 Target realization by gain adaptation

Once the flow-time profile γ_t has been fixed, the only remaining guidance parameter for generating x_n is the scalar gain w_n. Rather than choosing this gain independently of the desired outcome, we determine it from the prescribed target $y_n^\star$.

For a fixed guided atmospheric context

$$\left(x_{n-2}^{\text{gui}}, x_{n-1}^{\text{gui}}\right),$$

flow-time profile γ_t, and initial residual-noise realization $z_{n,0}$, running the complete residual flow defines the generated state as a function of the scalar gain,

$$x_n^{\text{gui}}(w_n). \tag{4.21}$$

Setting $w_n = 0$ removes guidance from the current residual flow while leaving its atmospheric conditioning context and stochastic realization unchanged. By the definition introduced in Section 4.1,

$$x_n^{\text{gui}}(0) = x_n^{\text{ung}|\text{gui}}. \tag{4.22}$$

Positive values of w_n instead modify the residual flow according to the fixed profile γ_t.

We define the terminal target gap as

$$\xi_n(w_n) := \mathcal{M}_\pi\left(x_n^{\text{gui}}(w_n)\right) - y_n^\star, \tag{4.23}$$

This terminal gap should be distinguished from the intermediate target gap $\xi_{n,t}$ in Equation (4.4). The latter is evaluated on the clean-state estimate during the residual flow and determines the local guidance gradient. In contrast, $\xi_n(w_n)$ is evaluated on the final generated weather state and measures whether the prescribed target has actually been realized.

Target realization therefore amounts to solving the scalar root-finding problem

$$\xi_n(w_n^\star) = 0. \tag{4.24}$$

The flow-time profile γ_t determines the relative placement of guidance within the residual flow, while $w_n^\star$ determines the overall gain required for that profile to realize the target.

4.3.1 Secant search

Evaluating $\xi_n(w_n)$ requires running the complete residual flow. Rather than differentiating the terminal target gap with respect to w_n, we solve Equation (4.24) using the secant method, which estimates a root from successive evaluations of the scalar function.

Given two previously evaluated gains $w_n^{(j-1)}$ and $w_n^{(j)}$, the next candidate is

$$w_n^{(j+1)} = w_n^{(j)} - \xi_n\left(w_n^{(j)}\right) \frac{w_n^{(j)} - w_n^{(j-1)}}{\xi_n\left(w_n^{(j)}\right) - \xi_n\left(w_n^{(j-1)}\right)}. \tag{4.25}$$

The first evaluation uses $w_n = 0$ and therefore produces $x_n^{\text{ung|gui}}$. A second evaluation uses a positive initial gain, after which the secant updates are applied iteratively.

All evaluations at weather step n use the same guided atmospheric context, the same flow-time profile, and the same initial residual-noise realization $z_{n,0}$. Consequently, the mapping

$$w_n \longmapsto \xi_n(w_n)$$

changes only through the guidance gain rather than through stochastic or conditioning differences between repeated flow evaluations. The local guidance gradients $g_{n,t}$ are nevertheless recomputed during every evaluation, since changing w_n changes the noisy states and clean-state estimates encountered along the flow.

To keep the search consistent with the direction of the prescribed intervention, we consider only non-negative gains. If an evaluated gain reaches the target within the prescribed tolerance, it is retained. Otherwise, if the search terminates without finding a root, we select the evaluated gain with the smallest absolute terminal gap,

$$w_n^\star := \arg\min_{w \in \mathcal{W}_n} |\xi_n(w)|, \tag{4.26}$$

where $\mathcal{W}_n$ denotes the set of gains evaluated during the search. The numerical details of the search are specified in the experimental setup.

This procedure is repeated while generating each guided weather state $n = 1, \ldots, H$. The resulting sequence

$$w_1^\star, \ldots, w_H^\star$$

adapts the overall guidance gain to the member-specific target at each weather step. For $n > H$, no target is imposed and the autoregressive forecast continues without guidance.

Chapter 5

Evaluation

This chapter evaluates the guidance method in two stages. We first compare different choices for the guidance-strength profile through a structured set of quantitative experiments and select the configuration used in the remainder of the thesis. With this configuration fixed, we then investigate how the method behaves under different scenario specifications, building an empirical understanding of how the requested target affects the resulting trajectories.

Throughout the experiments, we use a small set of diagnostics to verify that guidance behaves as intended and to identify anomalous runs before interpreting their results.

The first evaluation, T1, provides a systematic comparison of candidate guidance-strength profiles across multiple experimental settings. We assess each profile using a common set of complementary quantitative criteria and aggregate the resulting comparisons to identify a configuration that performs robustly across the considered settings.

In T2, we fix the selected guidance configuration and vary the parameters that define the scenario of interest. Rather than producing another standardized ranking, this analysis examines how changes in the specification affect the generated trajectories. Quantitative measures are used alongside the resulting fields and trajectories where they help characterize and explain the observed behavior.

The findings of these evaluations provide the basis for constructing and analyzing a complete high-temperature storyline in the following chapter.

5.1 T0: Experiment diagnostics

Before interpreting an experiment, we inspect a small set of diagnostics to verify that the guidance procedure behaved as intended. Specifically, we check

- whether each guided member reaches its prescribed target within a defined tolerance;
- how the targeted regional quantity evolves over the generative process, in particular for oscillatory or irregular convergence;
- whether the final guided field exhibits conspicuous spatial artifacts relative to the unguided field; and
- the spatial structure of the isolated guidance effect.

For the final diagnostic, we compare the guided state with its matched unguided-under-guided-context counterpart. Since x_n^{gui} and $x_n^{\text{ung|gui}}$ share the same conditioning states, deterministic forecast, and initial residual noise,

$$\Delta x_n^{\text{guidance}} := x_n^{\text{gui}} - x_n^{\text{ung|gui}} = r_n^{\text{gui}} - r_n^{\text{ung|gui}},$$

isolates the effect of guidance within the current residual flow.

These diagnostics provide a first check that an experiment is suitable for the more detailed evaluations that follow.

5.2 T1: Guidance-profile evaluation

We compare the candidate guidance-strength profiles through four complementary properties of the resulting guidance effect. Each metric is aggregated over ensemble members, guided weather steps, and the experiments included in the evaluation sweep. The resulting scores are combined in a leaderboard to compare the different flow-time profiles under a common set of criteria.

5.2.1 T1.1: Multivariate propagation of guidance

The first criterion measures how strongly an intervention targeted at regional surface temperature propagates to the remaining atmospheric state. We consider the guidance effect

$$\Delta x_n^{\text{guidance}} = x_n^{\text{gui}} - x_n^{\text{ung|gui}},$$

and quantify its magnitude both within the target region and over the complete spatial domain. The resulting masked and global push scores summarize how strongly the intervention extends beyond the directly targeted quantity.

5.2.2 T1.2: Deviation from the unguided trajectory

We next measure how far the guided state is displaced from its matched unguided-under-guided-context counterpart. The comparison is performed in a low-dimensional representation of the atmospheric state obtained through principal component analysis. The resulting distance provides a common measure of how strongly each guidance profile modifies the trajectory relative to the corresponding unguided generation.

5.2.3 T1.3: Spatial deviation from the target mask

The guidance objective is defined using a simple rectangular spatial mask. A guided response that reproduces this pattern directly would indicate a strong imprint of the prescribed objective on the generated field. We therefore compare the spatial distribution of the guidance effect with the target mask using total variation distance.

We compute this distance separately for the targeted surface-temperature field and across the complete atmospheric state. Larger values indicate a stronger departure from the imposed mask pattern.

5.2.4 T1.4: Guidance-effect diversity across ensemble members

Finally, we assess whether the spatial guidance effect remains dependent on the stochastic realization or converges toward a common perturbation across ensemble members. We quantify the spread of the guidance footprint across members, both for the target field and across all atmospheric fields.

Larger footprint spread indicates greater member-to-member diversity in the response to guidance.

5.3 T2: Scenario-specification analysis

Chapter 6

Evaluation

We evaluate the developed algorithm against the two requirements we have set for it:

- Quantitative evaluation.
  - Q1: Is it able to achieve the target we have set for it, across different experimental settings?
  - Q2: Which guidance schedule yields the most desirable results?
  - Q3: What results do the different experimental settings achieve, and what is their impact?
  - Q4: Are the results realistic in terms of spatial distribution, levels distribution, and co-variation, and time progression?

We organize the evaluation evaluation as follows. We answer the first two questions quantitaviley and proceed with a more qualitative approach for the latter two concerns. While address the third question with both quantitative metrics but with a qualitative approach, aimed only at characterize different settings. The last question would call for a quantitative approach, however, this would require both expert knowledge and a thorrow set of tests involving historical data, which is out of scope for this thesis. We thus address each one of the subquestions, but only assessing the visual characteristics.

6.1 Quantitative evaluation

6.1.1 Q2

6.1.2 Q3

6.2 Quanlitative evaluation

6.2.1 Q3

6.2.2 Q4

Here we assess the algorithm under different guidance schedules, with the purpose of selecting the most adequate parameters to perform the weather-time evaluation.

Before asking whether the method produces realistic results, we first have to establish that it works: that it reaches the target, does so reliably across settings, and does so in a desirable manner (stability and deviation concerns).

We test the achievement requirement and well–behaveheadness of the trajectory for different guidance algorithms through the following tests:

T1 – Reliability
: The gap to the target is driven to zero. We report the mean and standard deviation of the final gap.

T2 – Stability
: The trajectory is free of oscillatory behaviour. We measure this simply with respect to the pushback measured in the average trajectory (gui vs. ung steps) and whether the algorithm overshoots.

T3 – Closeness
: We check that the guided trajectory does not deviate excessively from the unguided one. We report the total amount of guidance needed, and the pathlength in PCA latent space, and the cosine similarity between the steps, compared to the gui-ung run.

TODO: set a bad initialization on purpose

6.3 Weather-time assessment

In weather-time we first characterise the parameters of the experimental setting—to describe their properties and to select settings for the larger experiments—and then validate the realism of the resulting trajectories. Throughout, we pay special attention to the diversity of the produced results as a secondary property of interest, alongside realism.

6.3.1 Parameter characterisation tests

In the following we provide a set of qualitative tests to better understand the role of the main parameters of the guidance at weather time. We test parameters in isolation, using the remaining parameters as controls. We proceed in three stages, examining in turn the region of interest, the variable of interest, and the intervention profile. For each we show different charts and metrics to expose what happens across these settings.

— maybe here is where the baseline would shine.

The reason for these tests is mostly to enhance our intuition about the role of each parameter, and provide a guideline on to select parameters for the large use cases we aim to produce with this work.

Region of interest

Here we assess variability of results by changing size, shift (from center), and coast-land covering of region of the interest.

Variable of interest

For simplicity we focus on heatwaves, and therefore on surface temperature. To probe the informativeness of the gradient, in this test we push the most correlated variables in an attempt to reproduce the pattern obtained by pushing temperature directly.

Intervention profile

Here we assess the degradation of the target patterns across intensity profiles. The intuition is that at some point things should brake down in some trivial way.

6.3.2 Realism tests

Once the guidance method and its parameters are selected, we assess the realism of the generated trajectories. Realism is central to this work, because we deliberately guide the generative model, possibly outside its training distribution. The tests themselves, however, are not specific to this work: they concern the general problem of evaluating the coherence of data-driven models, namely that the generated states should be consistent with the ground-truth dataset in both the guided and unguided settings.

We assess coherence to the ground-truth dataset along four dimensions:

T4 (Temporal).
Transitions of a variable across weather steps should be smooth.

T5 (Spatial).
Patterns over the region of interest should be spatially coherent.

T6 (Vertical).
Patterns should be coherent across pressure levels. We test this using a level rank coherence metric.

T7 (Cross-variable).
Patterns should be coherent with the learned weather transition function across variables. However, this is difficult to test quantitatively, we thus only make a qualitative assessment. (TODO: use the expert knowledge here)

We benchmark these properties against the ground-truth data as anchor (reference set), and against the unguided forecast as baseline, comparing unguided, guided-minus-unguided, and guided states. For T4–T6 we compute difference maps along the axis of interest and report the min, mean, max, and standard deviation across maps. For T7 we compute the cross-variable distributions over the weather steps of the ground-truth data, and report the deviation of the unguided, guided-minus-unguided, and guided states from their average.

For all four tests we average over the experiments and report the mean and standard deviation across runs. This indicates whether the generated extremes are faithful to the data distribution. A subtle but important point: for this to be meaningful, the base forecast must have missed the potential heat spots. The extreme is not only about the average—which the forecast most likely covers faithfully—but about the extreme spots, so both the mean and the max must be tested.

Chapter 7 sets out the concrete settings for each stage of this framework and reports the results, which Chapter 8 then discusses.

6.3.3 Method Validation

This approach can be used to guide any state towards an extreme over the specified mask.

To validate that the method is doing something sensible we can retrieve real heatwaves from the test dataset (everything that is not included in the train set), and observe what the model is doing.

Another way would be to guide it towards the ground truth and observe the RMSE of other variables compared to the ung.

Validate method against: gt, opposite direction, same direction. Make sure you apply same amount of guidance needed when defining y star.

Chapter 7

Experiments

TODO: before starting with the quantitative evaluation, start just by showing some guidance effect maps and absolute guided states with the rollout trajectories plot.

This chapter carries out the evaluation framework of Chapter 6. Each section follows the same pattern: we first state the concrete settings, then report the results and the decision they lead to. Tests are referenced by the identifiers introduced in Chapter 6 (T1–T7). —actually reference the latex identifiers like for equations

7.1 Flow-time evaluation

Note-taking:

- it seems like the gradient is most informative when we only evaluate it once (spike@ instead of spread), and we do it at the t=2. With informativeness we mean in terms of the push charts.

7.1.1 Setup

We first investigate how the placement of guidance within the generative flow affects target realization. To isolate the effect of flow time, we keep the target region and intensity fixed while varying the atmospheric starting state and stochastic realization. We use the following setup:

- one fixed target mask over Europe;
- three different atmospheric starting states;

- $N = 1$ weather step and $M = 3$ ensemble members;
- one target intensity of $+0.25\%$;
- three single-step guidance profiles: `spike@1`, `spike@3`, `spike@6`, `spike@12`, `spike@24`.

Each spike profile applies guidance at only one flow step, allowing early, intermediate, and late interventions to be compared directly. The guidance gain w_n is optimized independently for each experiment using the target-realization algorithm described in Section 4.3. The resulting design comprises $3 \times 3 \times 5 = 45$ guided experiments.

Given the results of this first test, we now test something related. The question is then not only during which phase of the guidance it's most appropriate to guide, but whether to spread the guidance over multiple steps would be benefitial. We test this by defining a_t to be interpolated between spike 1-3, 3-6, ...

Of course optimizing this would be most appropriate. What we are trying to do here is to observe empirically whether some properties emerge before defering future research to the development of an appropriate algorithm, which could use this information for initialization.

7.1.2 Achievement tests

The difficulty of tuning the first and second algorithm make them less suitable than the third one. Regarding algorithm 1, having a lower computational complexity is useless in practice if we then have to perform multiple costly runs. Regarding algorithm 2, having a desirability requirement encoded in the loss itself only complicate things. Tuning the optimization of this algorithms outweights the benefits, and induces a much higher cost in practical terms to the algorithm.

At later stages, the gradient becomes more similar to the mask (is more diluted, less sparse), which makes the guided pattern at the end of the flow be more spread.

Using a spike@1 approach is essentially equivalent to shifting the starting point of the flow trajectory (especially for the region over the mask) and let particle flow across the learned ODE.

What we observe is that the model doesn't necessarily pushes back against us. In contrary, when the method is tuned well, the model will do most of the pushing itself (look at this in percentage).

7.2 Weather-time evaluation

Throughout the weather-time experiments we vary the start state while fixing $N = 2$ and $M = 3$: this controls for the effects of $N = 1$ and for seed-specific effects, while keeping computational and memory requirements as low as possible.

Note that we will use the same experiments for the parameter characterization tests and for the realism tests.

— the realism tests could simply be used in each parameter subsection

7.2.1 Parameter characterisation tests

Region of interest

Variable of interest

Intervention profile

7.2.2 Realism tests

7.3 Use cases

For the final use cases we fix the parameters selected above and increase N and M, producing the experiments we had in mind when setting the goals of the project.

7.3.1 Europe 2020-06-01

7.3.2 Australia 2020-01-01

7.3.3 North America 2020-07-01

Chapter 8

Discussion

8.1 Gradient

The gradient has the unique property that it encapsulates what should change with respect to the learned weather function.

8.2 Bonus

Distance in PCA space.

8.3 Mask

We experimented with a gaussian mask, but realized in our experiments that the best thing to do is equal weighting inside the region. The correct way to define the region of interest is to choose a greater mask size instead of using some weighting scheme fading out of an epicenter. The mask size should be chosen to cover the entire area of interest, ensuring that all relevant data is included in the analysis. A larger mask size can help to reduce noise and improve the accuracy of the results, while a smaller mask size may exclude important information. It is important to carefully consider the appropriate mask size for each specific application to achieve optimal results. Either way we (most probably) loose the global coherence for the guided weather states, so we might as well define directly the region of interest with a mask. Additionally, this enables us to actually make statements about "the region of interest", having equal weights everywhere. Introducing different weigths might bias the generation in a way that is not desired.

We will see that including or not a specific area will change the guided patterns substantially. This however is consistent with the notion of "region of interest". If we consider a greater area for the changes, the changes will be applied respectively.

— note here: it may also be that the pushed area is coherent with the rest, and that the next deterministic prediction influences other regions too It is really important to understand how to evaluate what we are doing here. Since we are aiming at a specific region of interest, and we are guiding mainly locally over the region of interest, loosing the global coherence of the weather states. We shall thus include this in our evaluation by comparing inside region, outside region, and full spatial domain.

8.4 ...

The algorithm doesn't produce diverse results, because the gradients are really stable. This has to do with the fact that we have learned one and only one flow model (transition function). If we had multiple residual diffusion model, we would expect the possible effects to be more diverse. However, the second interpretation might be that the gradient provides with the information about the most likely direction of change (which is true under the learned data distribution).

This happens because the guidance vector is a scaled version of the flow model's gradient wrt to the noisy state. The gradient is stable across the full trajectory, because the only variable inputs are the noise itself and the timestep of the flow. The stochasticity of the noise is not enough to produce diverse results. The direction of the gradient is always the same. If we say "let the masked average go up by phi percent", the only thing that changes is the magnitude of the gradient, but not its direction, which is specified by the Jacobian.

This is in contrast to guidance settings.

Shift between atmosphere and stratosphere.

Put here all observations

Chapter 9

Conclusion

9.1 Summary

9.2 Future Directions

9.3 Conclusive Remarks

Appendices

Appendix A

Development of the target-realization algorithm

Before arriving at the target-realization algorithm presented in Section 4.3, we explored two more flexible strategies for determining the guidance magnitude: Close-Gap and Optimize-Schedule. These preliminary approaches motivated the final choice to prescribe the relative flow-time profile and optimize only a single overall guidance gain.

A.1 Close-Gap

The first approach exploits the fact that the desired target is known before generation by prescribing how the remaining target gap should evolve throughout the flow. Let

$$\bar{\xi}_{n,t} := \gamma_t^{\text{gap}} \, \xi_{n,0}, \tag{A.1}$$

denote the desired remaining gap at flow step t, where γ_t^{gap} specifies the fraction of the initial gap that should remain.

At each flow step, the method computes the local displacement of the noisy state required to move the current gap $\xi_{n,t}$ towards $\bar{\xi}_{n,t}$. Using a first-order Taylor approximation,

$$\xi_{n,t}(z_{n,t} + \Delta z_{n,t}) \approx \xi_{n,t}(z_{n,t}) + \left(\nabla_{z_{n,t}} \xi_{n,t}\right)^{\top} \Delta z_{n,t}. \tag{A.2}$$

Requiring the corrected gap to equal the prescribed value gives

$$\left(\nabla_{z_{n,t}}\xi_{n,t}\right)^{\top}\Delta z_{n,t}=\bar{\xi}_{n,t}-\xi_{n,t}. \tag{A.3}$$

Among the displacements satisfying this local constraint, the smallest in Euclidean norm is obtained by moving along the gradient of the gap,

$$\Delta z_{n,t}^{\star}=\frac{\bar{\xi}_{n,t}-\xi_{n,t}}{\left\|\nabla_{z_{n,t}}\xi_{n,t}\right\|_2^2}\nabla_{z_{n,t}}\xi_{n,t}. \tag{A.4}$$

Using

$$g_{n,t}=\nabla_{z_{n,t}}\xi_{n,t}^2=2\xi_{n,t}\nabla_{z_{n,t}}\xi_{n,t}, \tag{A.5}$$

the desired displacement can equivalently be written as

$$\Delta z_{n,t}^{\star}=-\frac{2\xi_{n,t}\left(\xi_{n,t}-\bar{\xi}_{n,t}\right)}{\|g_{n,t}\|_2^2}g_{n,t}. \tag{A.6}$$

The guidance contribution to the Euler update is

$$\Delta z_{n,t}^{\mathrm{gui}}=-h_t\lambda_{n,t}g_{n,t}. \tag{A.7}$$

Requiring this contribution to realize the desired local displacement yields

$$\lambda_{n,t}^{\mathrm{CG}}=\frac{2\xi_{n,t}\left(\xi_{n,t}-\bar{\xi}_{n,t}\right)}{h_t\|g_{n,t}\|_2^2}. \tag{A.8}$$

The attraction of this formulation is its efficiency: the guidance magnitude is determined online and only a single flow is required. However, the update is greedy. It computes the intervention required to satisfy the prescribed gap under a local approximation, without accounting for the subsequent displacement induced by the learned velocity field. The resulting trajectory can therefore overshoot the prescribed gap and require compensating corrections at later flow steps.

A slower gap-closing profile can reduce this behaviour, but then introduces a schedule parameter controlling how aggressively the gap is closed. Choosing this parameter requires repeated runs, reducing the practical advantage of the otherwise single-pass method. For this reason, we next considered optimizing the flow-time intervention schedule directly.

A.2 Optimize-Schedule

Rather than prescribing the evolution of the target gap, Optimize-Schedule treats the intervention magnitudes across the complete flow as optimization variables.

We parameterize the intervention through its contribution to the Euler update,

$$k_{n,t} := h_t \lambda_{n,t}, \tag{A.9}$$

such that

$$z_{n,t+1} = z_{n,t} + h_t u_{n,t} - k_{n,t} g_{n,t}. \tag{A.10}$$

The quantity $k_{n,t} g_{n,t}$ is the state displacement induced directly by guidance at flow step t. We optimize the complete intervention trajectory

$$\mathbf{k}_n = (k_{n,0}, \ldots, k_{n,T-2}) \tag{A.11}$$

using the objective

$$J_n(\mathbf{k}_n) = \xi_n^2 + \kappa \sum_{t=0}^{T-2} \|k_{n,t} g_{n,t}\|_2^2. \tag{A.12}$$

The first term measures target realization, while the second penalizes the total squared magnitude of the guidance-induced state displacements. This allows the optimization to prefer schedules that approach the prescribed target while limiting the amount of intervention applied along the flow.

In principle, this formulation removes the need to prescribe a gap-closing trajectory and allows the temporal distribution of guidance to adapt freely. In practice, however, the optimization is strongly coupled. Changing $k_{n,t}$ modifies the noisy trajectory, which changes the clean-state estimates and therefore also the guidance gradients encountered at subsequent flow steps. Consequently, the optimal intervention at one flow step depends on the interventions applied throughout the trajectory.

In preliminary experiments, this high-dimensional optimization was strongly sensitive to its initialization and could converge rapidly to different local solutions. The additional flexibility therefore came at the cost of a substantially more difficult optimization problem.

These observations motivated the lower-dimensional formulation adopted in the main method.

Instead of determining an independent intervention magnitude at every flow step, we prescribe a normalized profile γ_t and restrict the guidance schedule to

$$\lambda_{n,t} = w_n \gamma_t. \tag{A.13}$$

Target realization then reduces to optimizing the single scalar w_n at each weather step, as described in Section 4.3.

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